

Play the game then forever can begin: Designing games to extend gameplay

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ABSTRACT

Online games are among the most popular digital information systems for entertainment. Game providers frequently encounter the problem of retaining players. Players' expressed interest in gameplay may not always be transformed into persistent gameplay behavior, posing challenges for game providers to extend players' actual gameplay. To address this problem, we introduce a new concept, gameplay extension (measured in terms of the number of months a player actually continues playing a game), and explore how game elements can be designed to enhance gameplay extension. Drawing on goal gradient theory, we also examined the possible interplay of game design with gaming skills to formulate gameplay extension. We tested our model using multiwave and multisource data, including psychological and system-captured gameplay data from 379 players. We found that the mechanism affordance positively affects goal progress and concentration. Interestingly, concentration has a stronger effect than does goal progress on gameplay extension. Moreover, our findings uniquely clarify the pathway through which role-playing affordance extends gameplay, particularly among low-skilled players. Overall, our model explained 41% of the variance in gameplay extension, demonstrating its practical relevance. We argue that the findings apply to player-concentrated games or gamified systems.

1. Introduction

Online games are a highly prevalent IT application and one of the most engaging forms of entertainment worldwide. The global gaming market is estimated to surpass 503 billion US dollars in 2025 (Statista, 2025a), while the number of global online game players is projected to reach 3.02 billion (Statista, 2025b), and more than 5000 game manufacturers worldwide have participated in the online game market (Gamedevmap, 2025). In such a highly competitive environment, game companies urgently need to understand how to maintain players' engagement in gameplay for a substantial time to increase revenue potential, warranting research into how to increase players' gameplay extension, which we define as the number of months a player actually continues playing a game, with reference to Huang et al. (2024a).

Past game studies have examined a number of antecedents to gameplay extension in terms of player loyalty, continuance intention, and actual use: reward responsiveness, fun seeking (Aydinlyurt et al., 2021), cooperation knowledge (Bian & Yang, 2025), flow, and engagement (Rahman et al., 2024). However, it is not yet fully known how a game can be designed to motivate players to move beyond an intention to increase their gameplay over extended periods, indicating a research gap. Research addressing this topic should make practical and scholarly contributions. In practice, such research helps game providers make informed decisions that extend their players' actual gameplay. Academically, such research opens a new methodological path, i.e., a valuable approach for tracing players' actual game use over months.

To address this gap, we draw on goal gradient theory (GGT), according to which individuals make greater efforts toward achieving a

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goal the closer they are to it (Hull, 1932; Kivetz et al., 2006). GGT can be used to address this gap, as players' continued engagement is often motivated by the goal-pursuit found in games, although previous research has mostly ignored such goal designs in a game. GGT must have enhanced applicability in our study because game players have strong goal-attaining motivations and enjoy the associated achievements (Huang et al., 2022; Lin et al., 2015). We used GGT to identify the process variables. First, GGT indicates individuals' tendency to expend significant effort on goal pursuit (Hull, 1932; Kivetz et al., 2006). Effort spent should comprise cognitive resources focused on the focal activity, i.e., *concentration* (Gaillard, 2018), justifying its inclusion as an important process variable. Second, GGT research verifies that individuals are highly motivated as they perceive themselves moving closer to achieving their goals (Förster et al., 1998), i.e., *goal progress* (Teng et al., 2022); thus, we include it as the second process variable, as the game rules in most games, if not all, require players to work until their goals are met.

We further used GGT and the two process variables to identify important design elements. Design elements can be effective, as they can assist users' goal pursuit, i.e., by providing affordances (Strauss et al., 2024; Teng et al., 2024). GGT has two theoretical components: end goals and deeming goals "larger" (Förster et al., 1998). End goals are tied to experiencing the ends of the storylines of in-game characters that comprise our definition of *role-playing affordance* with reference to Hollebeek et al. (2022). Deeming goals "larger" requires the knowledge on clear understanding of in-game rules and numbers, thus enabling optimization along the pathways toward attaining goals, which we define as enhanced *mechanism affordance* with reference to Kim (2021).

Thus, our research examined how the two affordances (mechanism and role-playing) affected goal pursuit (concentration and goal progress) and eventually improved gameplay extension (i.e., the number of months out of four a player continued to play the game from the end of the second data survey). While skills have been considered important for accomplishing tasks or goals (Przegalinska et al., 2025), they may change players' abilities in actualizing game affordances, thus affecting their effect on gameplay extension and therefore motivating us to further examine the moderating role of skill levels. Our research achieved this by using data collected from multiple sources and at multiple time points. This approach has methodological merit. The results indicate that not all affordances motivate the two aspects of goal pursuit. Moreover, concentration better fosters gameplay extension than goal progress does. Role-playing affordance indeed has a reduced effect on gameplay extension for players who have high skill levels.

Overall, our findings answered two questions: how can games be designed to motivate goal pursuit and further actual use for months, and how gaming skill level can change the answer to the first question? Specifically, our findings make several contributions. First, the findings uniquely indicate that the mechanism affordance can effectively focus players' attention on gameplay and, therefore, enhance gameplay extension, offering a novel pathway from game design elements to gameplay extension. Second, gameplay extension was conceptualized using consecutive months of actual gameplay, which is new to the literature but highly relevant to practitioners. Third, the findings extend the GGT literature by indicating that concentration may be more effective than goal progress in terms of user retention in the online game context. Fourth, our findings indicate that the effective design of game affordances should also consider player skill levels. Overall, 41% of the variance in gameplay extension is explained by our model, showing the practical relevance of our research.

2. Theory and prior research

2.1. Game affordances

Affordances are action possibilities offered by technologies to enable users to achieve their goals (Strauss et al., 2024). In online games, game

affordances are inherent design elements within games that inform players' action possibilities (Lee et al., 2021) and meaningfully influence the cognitive abilities of players (Tarafdar et al., 2024). The concept of an affordance can be used to understand how players perceive and engage with game elements. Affordances encourage players to understand game mechanisms to achieve goals and progress through game narratives (Lee et al., 2021). Therefore, the influence of game affordances on the player experience is substantial and multifaceted.

The game affordances can be synthesized into three main affordances: achievement, social, and immersion affordances (Lee et al., 2021). Regarding in-game achievements, gameplay motivations may include advancement, competition, and mechanisms (Yee, 2006). Specifically, advancement could result in in-game progress, while the progress may be accumulation of in-game resources or avatars' items (Tai et al., 2025), while not attaining in-game goals. Competition is rather inter-player (Lin et al., 2015), while not necessarily intra-player goals, i.e., players may not set a goal of beating others. Mechanism indicates in-game underlying rules and numbers that enable players to optimize their gameplay as much as possible (Lee et al., 2021), while achieving their goals and thereby supporting its suitability. Regarding in-game socialization, gameplay motivations include socializing, relationship, and teamwork (Yee, 2006). All these are based on inter-player interactions that can either be positive or negative to goal pursuit, e.g., cheating and hoarding of advantageous locations (Teng et al., 2012), thereby suggesting instability in goal pursuit. Regarding immersion in games, gameplay motivations include discovery, customizing, escapism, and role-playing (Yee, 2006). Discovery of hidden things can be valuable to players (Tseng & Teng, 2015), but the hidden things may not be powerful, thus not necessarily be goal-driven. Customization meets individuals' needs to express themselves (Sundar et al., 2012), but expressing one's self unlikely helps achieve in-game goals. Escapism indicates game engagement to take a break from real-world stressors (Huang et al., 2024b). In this case, feeling relaxation is important (Huang et al., 2024b), which should not be substantially related to goal pursuit. Eventually, role-playing enables players to take the roles of the in-game characters and follow the storylines of the characters to reach the story ends that are closely tied to the in-game goals, showing the importance of role-playing affordance in gaming goal pursuit.

Among these affordances, mechanism and role-playing affordances uniquely contribute to goal-driven intention (as gameplay extension in our model). The mechanism affordance enables players to understand and utilize the underlying rules and systems, thus facilitating players' goal pursuit. The role-playing affordance enables players to better assume the in-game characters and fully follow the characters' storylines to reach story ends, thus fostering players' goal pursuit. Comparatively, immersion affordances literally immerse players in the game world, instilling a feeling that they are actually in the game world. Similarly, such a feeling may not necessarily be related to goal pursuit. For example, players immersed in the game world may simply experience its fantasy or just take a break from the real world, not necessarily motivating players to attain in-game goals. Moreover, other continuance mechanisms (e.g., social influence and habits) may not always foster goal pursuit. Specifically, social influence and its enabling affordances may introduce both positive effects (helping pursue the goals) and negative effects from other players (bullying and in-game impolite communication), not definitely helping goal pursuit. Furthermore, habits induce spontaneous engagement, but such engagement may or may not always be goal-driven. For example, habits may simply induce players to revisit the game to check its updates or wander the game world that are not necessarily goal-driven. In short, not all the affordances and continuance mechanisms (e.g., habit and social influence) may motivate goal-driven engagement. Therefore, our study chose to focus on mechanism affordance and role-playing affordance.

Game affordances offer players hedonic and psychological benefits, improving their satisfaction and enjoyment (Alalwan, 2020; Alalwan et al., 2017; Gong et al., 2020). Past studies have explained the

association between psychological constructs and goal pursuit (Hamari et al., 2020). Players try to pursue gaming goals (Teng et al., 2024) and feel satisfied and experience increased enjoyment when they gain the value of achievement, a hedonic benefit (Lin et al., 2015). However, the association between the fundamental nature of a game and goal pursuit is rarely mentioned in the literature, revealing a knowledge gap. To address this issue, we propose exploring games' fundamental properties, i.e., game affordances.

Players' skills are their abilities to attain in-game goals (Huang et al., 2022). Such abilities should accelerate players' progress toward attaining goals, thus changing the effects of affordances on attaining in-game goals. Specifically, high-skill players can use their understanding of in-game mechanisms (or the rules and the numbers) to draft better strategies. During gameplay, high-skill players can more easily modify their strategies to better ones, thus *better* transforming mechanism affordance to escalate their likelihood of attaining goals. Moreover, high-skill players can more easily use their abilities to overcome in-game obstacles. Therefore, even if they do not assume the role of the gaming characters and do not follow their storylines, they still have superior skills so that they play the game in an exhaustive manner to achieve the end goals, thus *weakening* the effect of role-playing affordance on attaining goals. In summary, skills can moderate the effects of the two selected affordances.

2.2. Goal gradient theory

We use GGT as the theoretical foundation. GGT posits that individuals exert considerable effort toward achieving goals when they feel close to attaining their goals (Hull, 1932; Kivetz et al., 2006). GGT was initially proposed as a hypothesis and tested in animals. Specifically, rats were found to accelerate when they approach their goals (Hull, 1932). Such acceleration should have dynamics that the goals are more and more sensible along the last mile toward attaining them. GGT was applied in the marketing field to explain consumers' goal valuation in a short temporal distance (Song & Ko, 2014) and that they accelerate purchases when they are closer to the reward goals (Kivetz et al., 2006). GGT has also been applied to explain why employees approaching their reward goals work harder and, therefore, perform better (Camilleri et al., 2023). These findings support the concept that GGT is a robust theory that can explain human behaviors in various goal-pursuit contexts.

GGT has also been applied in online games. Players who expect that their avatars will grow in the near future will encourage themselves to pursue gaming goals (Teng et al., 2022). Moreover, players' expectations that their gaming skills will improve have a similar effect on motivating players to pursue their gaming goals (Huang et al., 2022). Furthermore, players who perceive that they have long, deep, and broad relationships with the game feel that they are close to their gaming goals, thus strengthening their game loyalty (Teng, 2017a). These applications show that the GGT explains players' goal pursuit behaviors. However, these studies have not yet included game affordances to encourage players' goal pursuit behaviors, indicating a research gap. The present study fills this gap by introducing mechanism affordance and role-playing affordance as novel and important instruments that drive players' goal pursuit.

GGT can be used to theoretically identify two important affordances: the mechanism affordance and role-playing affordance. Specifically, GGT has two important components: end goals and deeming goals "larger" (Förster et al., 1998). First, end goals in games are typically story ends of the in-game characters. Hence, players fully assuming the roles of the in-game characters and closely following their storylines are more likely to reach the story ends, i.e., perceiving enhanced role-playing affordance. Second, deeming goals larger in games should be a perception that the goals are easier or at least have a clear pathway toward attaining them. Clear pathways can be built when players clearly know the underlying game rules and numbers, i.e., perceiving enhanced

mechanism affordance.

GGT includes elements related to goal pursuit: goal proximity and motivation to attain goals (Teng et al., 2022). However, to increase both elements, individuals must first exhibit *goal progress*, i.e., moving toward their goals. Goal progress can motivate individuals to exert significant effort (Kivetz et al., 2006), which likely reinforces goal pursuit, thus theoretically justifying the inclusion of goal progress in our research. Goal progress can give individuals a sense of being closer to the goal; therefore, the GGT predicts that such individuals would exert significant cognitive effort (Förster et al., 1998), i.e., *concentration*, theoretically justifying its inclusion.

Concentration does not imply any particular skill level and, therefore, differs from flow. Flow requires an increased skill level (Huang et al., 2022), showing how it is distinct from concentration. Concentration also differs from immersion since it does not require a sense of "being there," which defines immersion.

Concentration (or focused attention) should result in better learning and performance (Gaillard, 2018), both of which provide positive feedback that may trigger continuous engagement, supporting our model that concentration increases gameplay extension. Moreover, GGT can be used to infer that goal progress strongly motivates players to further engage in gameplay, i.e., motivating repeated gameplay, thereby increasing gameplay extension, justifying our theoretical link from goal progress to gameplay extension.

Past research has examined gameplay extension using continuance intention (Hamari et al., 2020), loyalty (Bian & Yang, 2025; Pham et al., 2022), and the total number of game sessions in a given time period (Huang et al., 2024a, 2025b). Nevertheless, these studies have not examined gameplay extension in terms of how many months players continue to play a game, indicating a research gap. The present research addressed this gap using a month-based construct of gameplay extension and operationalized it through system-generated data. Gameplay extension in our study differs from actual continuance behavior: the former includes *how many months* of continuance behavior, while the latter may simply concern *whether* there is continuance behavior. Moreover, gameplay extension differs from the objective gameplay measures in the past research, i.e., Huang et al., (2024a), (2025b): the former counts how many *months* of gameplay (reflecting extended gaming history), while the latter counts how many *sessions* of gameplay (reflecting gaming frequency). We chose to use gameplay extension, as it better captures customer retention concept, which game providers should concern more.

Gaming skills help players make in-game progress toward achieving goals (Huang et al., 2022). Hence, gaming skills can be expected to increase goal proximity, an important element in GGT. Moreover, gaming skills should provide players confidence (or self-efficacy) in attaining goals. Increased self-efficacy boosts action motivation (Teng et al., 2022). Hence, gaming skills should be related to motivation to pursue gaming goals, another important element in GGT. Collectively, gaming skills can be integrated into GGT literature and therefore are suitable for inclusion in the research model. Table 1 presents and defines our

Table 1
Study Concepts and Definitions.

Concept	Definition
Mechanism Affordance	The possibility perceived by a player to understand the precise number and percentages behind the in-game computation
Role-Playing Affordance	The possibility perceived by a player to experience the in-game characters' stories
Concentration	The percentage of gameplay time during which a player feels fully focused
Goal Progress	A player's perception of moving toward their in-game goals
Skill Level	A player's officially recognized gameplay ability
Gameplay Extension	The number of months out of four in which a player continues to play the game from a particular time point

research concepts.

We built the study model by drawing on GGT. This model is new in four ways. First, it presents a novel conceptualization of goal pursuit comprising two elements: concentration and goal progress. Second, the two game affordances are uniquely identified as the means to fuel these two elements. Third, it is new to consider skill level as a moderator of the effects of game affordances. Fourth, gameplay extension is operationalized with system-captured behavioral data.

The novelty of this model also provides practical insights. The testing results of this model could guide game designers to appropriately use their resources to improve the affordances that best support the influential pathway (concentration or goal progress). Fig. 1 presents the study's model.

Note. Ellipses indicate items assessed using survey responses. Rectangles indicate concepts measured using system-captured data. The control variables were modeled to have effects on all the endogenous variables.

Players' understanding of in-game mechanisms is a crucial component of their motivation for gameplay, which reflects their desire to optimize their characters' in-game performance (Yee, 2006). Hence, players appreciate games that afford them the opportunity to understand the in-game mechanisms. For example, in the game Valorant, players can learn about the economic system in the weapons purchase interface, which determines the currency that can be obtained in the next round; this is highly important tool for ensuring victory, and making corresponding game strategies (ExitLag, 2024). The mechanism affordance enables players to learn a game's fundamental operating principles (Kim, 2021). Such knowledge can help players understand in-game mechanisms and how their gameplay can lead to favorable outcomes. That is, the mechanism affordance helps players design better game strategies and thereby attain in-game achievements (Jang & Byon, 2020). Better strategies give players confidence in achieving their goals. According to GGT (Huang et al., 2022), they should be encouraged to focus their attention on attaining close goals, i.e., boosting concentration.

H1a. : The mechanism affordance has a positive association with concentration.

Role-playing is an important experience during gameplay (Hollebeek et al., 2022). The role-playing affordance indicates how a game enables players to follow the storylines of the gaming characters (Lee et al., 2021), thereby satisfying their gaming motivation to role-play game characters (Yee, 2006). In online games, role-playing affordances encourage players to experience the stories of gaming characters and

design their own characters. For example, players can select different clothing and badges during gameplay (Aldemir et al., 2018). In so doing, players feel that they are represented by the gaming characters (Hollebeek et al., 2022). Hence, the goals of the gaming characters are their own goals. According to GGT (Kivetz et al., 2006), because players are strongly motivated to reach their own goals, they concentrate their attention on the game content.

H1b. The role-playing affordance has a positive association with concentration.

The mechanism affordance reflects how a game enables players to know and analyze a game's computations and operations. Such action possibilities motivate players to explore the underlying principles and rules of the game, thereby enabling them to develop game strategies that enhance their characters' performance (Kim, 2021). One typical performance measure is the reduced distance toward attaining in-game goals. Hence, mechanism affordance instills players a perception of being closer to attaining their goals, which in GGT literature (Teng et al., 2022) was conceptualized as goal progress, supporting the rationale of this hypothesis. Moreover, a deeper understanding of game mechanisms empowers players to formulate advanced game tactics, which can improve their self-efficacy (Bian & Yang, 2025) and lead to in-game achievements (Wang et al., 2021). Enhanced character performance and in-game achievements facilitate meaningful progress toward in-game goals (Jang & Byon, 2020).

H2a. The mechanism affordance has a positive association with goal progress.

The role-playing affordance is how a game enables players to experience the fantasy world by using in-game avatars, storylines and avatar histories, thereby fulfilling players' motivation for role-playing (Yee, 2006). In the online game Monster Hunter: World, the character creation system allows players to adjust the avatar's hair, tone of voice, makeup, and other details and to see the created avatars in subsequent plots (GameFragger, 2025). By actualizing role-playing affordances, players can role-play their own avatars and experience the storylines. During the experience of storylines, players are getting closer to the story ends of the gaming avatars. In the GGT literature, the closeness to the goals is defined as goal progress (Teng et al., 2022), supporting the rationale of this hypothesis. Moreover, role-playing avatars instill in players a sense that they are in the mediated environment, perceiving high levels of avatar-self relevance (Hollebeek et al., 2022), which fits the definition of avatar identification (Teng et al., 2023). Avatar identification motivates players to further engage in gameplay, progressing toward

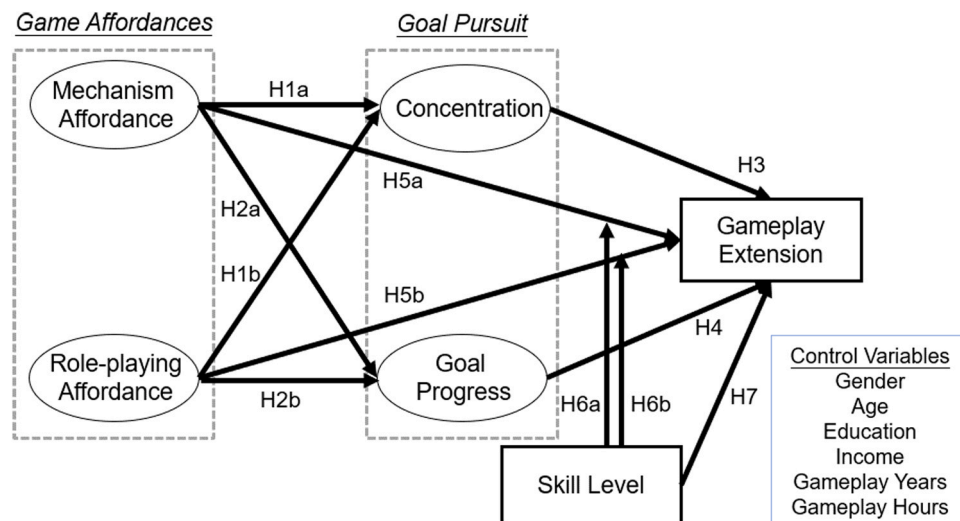


Fig. 1. Study Model.

in-game goals (Pham et al., 2022). That is, the role-playing affordance can facilitate substantial goal progress by players.

H2b. The role-playing affordance has a positive association with goal progress.

Concentration indicates devoting attentional resources to a task; that is, players are focused on gameplay and therefore experience improved learning and performance (Sadiyani, 2018). Hence, concentration leads players to feel they are approaching their goals. According to GGT, approaching goals motivates players to exert more effort (Kivetz et al., 2006). Greater player effort in gameplay often translates into increased time investment, such as sustained engagement over several months, which is the core of the definition of gameplay extension.

H3. Concentration has a positive association with gameplay extension.

In the online gaming context, players are motivated to attain goals (Yee, 2006), for example, by following leaderboards and measuring other players' progress against their own. Stronger motivations to attain gaming goals drive players to exert more effort (e.g., time) on online gameplay (Teng, 2017a), allowing them to become closer to reaching the game goals. According to GGT, players who are closer to achieving their goals see those goals as larger (Förster et al., 1998), indicating a greater likelihood of achieving them. In-game achievements produce high satisfaction (Xi & Hamari, 2019), leading to game continuance (Aydinlyurt et al., 2021), which can increase the number of months spent playing a game (Hyeong et al., 2020). As gameplay extension is defined as the number of months the nominated game was played, we hypothesize the following:

H4. Goal progress has a positive association with gameplay extension.

Enhanced mechanism affordance reflects abundant information about game rules and numbers (Kim, 2021). Such abundant information may evoke curiosity of players to experience the numbers and rules to see what they are, how they feel, and whether players can take advantage of them. Curiosity motivates further engagement, e.g., customizing avatars and increasing purchases (Ahn et al., 2024). An enhanced game mechanism comprises a huge amount of information, which can be explored by players for many months; that is, increasing their gameplay extension. Therefore, we hypothesize:

H5a. Mechanism affordance has a positive association with gameplay extension.

Role-playing indicates that players assume the role of game characters and experience their storylines (Hollebeek et al., 2022; Yee, 2006). By moving forward along the storylines, players experience what happen to the game characters step by step. Gradually, players can develop an emotional attachment to the game characters or even feel that they are the game characters in the game world. Both phenomena can increase further engagement (Tai et al., 2025; Teng, 2017b). Applied in the game context, increased action possibility of role-playing (i.e., role-playing affordance) can increase players' game engagement, thus extending their gameplay for months. Therefore, we hypothesize:

H5b. Role-playing affordance has a positive association with gameplay extension.

The mechanism affordance represents the within-game computational process, which consists of players analyzing the underlying rules and systems (Yee, 2006). Knowing the within-game computational process, or mechanism, helps facilitate the transformation of gameplay into favorable outcomes. Specifically, the mechanism affordance enables players to understand the game rules. Fully understanding the game rules inspires players to adjust their gameplay strategies to win the game (Jang & Byon, 2020). The effects of mechanism affordance may differ among high-skill and low-skill players, as skills help players complete tasks (Przegalinska et al., 2025) and therefore skills can help actualize affordances. High-skill players can better utilize their

knowledge of the game rules to design good strategies, resulting in better gaming outcomes and therefore satisfying their competence needs. Competence need satisfaction induces individuals' repeated engagement (Deci & Ryan, 1987) and encourages repeated use (Lin et al., 2015), thus increasing gameplay extension. In contrast, low skill levels restrains players from effectively using knowledge of game mechanisms to devise good strategies, or fail to competently implement such strategies to obtain good gaming outcomes; these scenarios are likely to reduce gameplay extension. Hence, we hypothesize:

H6a. Skill level strengthens the link between mechanism affordance and gameplay extension.

The role-playing affordance refers to how a game enables players to assume the role of game characters and experience their storylines. That is, role-playing affordances can fulfill players' motivation to role-play, a major motivation (Yee, 2006). In online games, players can feel inspired to use the in-game characters they design to engage in gameplay (Teng et al., 2024). Moreover, players can change their characters' appearance repeatedly to increase their affective engagement (Brown et al., 2024). Therefore, high affective engagement inspires players to devote time and effort to experience their characters' storylines, thus increasing their gameplay extension for attaining achievements. Such effects of role-playing affordance may differ between high-skill and low-skill players. High-skill players can obtain game achievements with or without affective engagement, with or without following the characters' storylines. In contrast, low-skill players need to follow the storylines to obtain more cues and ideas for making in-game progress and for increasing the affective engagement. Both foster the gameplay extension. Hence, we hypothesize:

H6b. Skill level weakens the link between role-play affordance and gameplay extension.

Skill level is defined as the officially recognized gameplay ability of a player. A high level of skill denotes marked in-game improvement, further contributing to the player community (Liao et al., 2024). In the game World of Warcraft, avatars need to advance their levels to obtain a lock-picking skill, enabling them to open a treasure chest or door (Wowpedia, 2025). High skill levels make goals easier to attain or seem closer and, according to GGT (Förster et al., 1998), motivate further engagement. High levels of gameplay ability give players a sense of self-efficacy, which can fuel playing games (Teng et al., 2022) and using other online applications (Loh et al., 2022). Superior gaming skills further enable players to meet others' expectations, particularly when they perceive other avatars as real persons (Huang et al., 2025a). A high level of gaming skill represents competent engagement in strategic planning and implementation during gameplay, supporting the expectations of the competence needs of other players in the same group to overcome difficult challenges (Cengiz et al., 2025; Hogg & Turner, 1987). Competence is important for building assurance (Yuan et al., 2024). Hence, satisfying the need for competence leads to individuals' repeated engagement (Deci & Ryan, 1987), i.e., repeated gameplay (Teng et al., 2024). Repeated use extends gameplay.

H7. Skill level has a positive association with gameplay extension.

3. Method

3.1. Data collection and participants

Our study employed a three-wave design. This design ensures the temporal sequences of the measured variables. Our study collected data using two means. The first-wave data were collected using an online survey between April 13 and May 12, 2021. In the second wave, the respondents were followed up between May 13 and June 11, 2021. For those who offered valid responses to both waves of online surveys, we checked their system-captured game use data in each of the subsequent

four months, comprising the third wave of data and enabling us to compute our dependent variable: gameplay extension. This design has three merits: (1) multiple waves, (2) multiple sources, and (3) the inclusion of consecutive objective data.

To solicit participation, we posted invitations on various websites and forums, such as Instagram, Facebook, ptt.cc, and gamer.com.tw. Including various websites and forums aimed at reaching multiple respondent sources, thus reducing any bias due to a single respondent source. The stated research purpose was to understand players' perceptions of online games and their in-game goals. Any interested respondent clicking on the supplied link was directed to the survey form, which was powered by SurveyCake, a locally popular online survey service. The first item of the survey solicited the respondents' consent for participation, completing the informed consent process. The survey's introduction stated that any respondent who returned complete responses would be entered into a drawing in which 50 respondents would obtain a coupon valued at NT\$200, i.e., US\$7.10. Obtaining this gift is probabilistic; therefore, the expected value should be lower, resulting in a moderate expected value. A moderate reward may not result in self-selection response bias (Jia et al., 2017).

During the second wave, this research followed up with 1853 respondents who offered complete and valid responses. The valid responses were determined using the following criteria: (1) nominating an existing online game; (2) indicating nonzero gameplay time in the month before the survey; (3) indicating weekly play of the nominated game < 168 h (one week has a total of 168 h only); (4) being aged 18 years or over; (5) offering a nonduplicate email address; and (6) passing the attention check. Additionally, we filtered out respondents who did not provide complete demographic information and excluded those who reported the highest 1% value (as outliers) in gameplay years. We believe that these criteria can rule out insincere and unserious responses. The sample size may be deemed large. This may be a result of a large local player population and trust in the investigators who work in reputable local research institutes. We further asked those players who nominated League of Legends (LoL, a popular online game) to provide their game account names. In total, we obtained 379 respondents' account names, which enabled us to check their system-captured gameplay data.

We chose LoL, a five-versus-five competitive game, to implement our study. The reasons are as follows. First, LoL records detailed gaming outcomes, e.g., kills, damage outputs, win rates, and accumulated gold (in-game currency), which can be players' personal goals and used to analyze their game strategy. Therefore, players are motivated to improve their win rate by understanding the mechanisms of the game. Second, each player must choose their champion avatar to play the game, e.g., Mage, Archer, or Assassin. Each avatar has unique in-game abilities, e.g., a Mage can inflict considerable damage on enemies, and an Archer can heal team members. By playing with different avatars, players can have multiple role-playing experiences. LoL features group gameplay; therefore, players' concentration on their team roles is critical to group gaming outcomes. These features conform to the study concepts in our research model, thereby justifying the appropriateness of LoL for our study.

Online surveys can be limited by nonresponse bias (Steelman et al., 2014), i.e., how many invitations viewers did not click on or complete the surveys. Hence, we adopted the second-best approach, i.e., comparing early and late responses, because late respondents may resemble those who did not respond (Armstrong & Overton, 1977). Accordingly, the present research compared the first and last quarters of the responses. No significant difference in the research concepts ($p > .64$) was shown; thus, nonresponse bias was not a significant concern.

3.2. Measurement

3.2.1. First-wave research variables and their development

We developed scales for measuring the two affordances. We collected items from the literature (Lee et al., 2021; Yee, 2006). We also invited three experienced scholars and two language experts from a government-certified agency to implement the process of forward and backward translation, thus ensuring semantic equivalence between the different language versions. We implemented a content validity assessment conducted by five faculty members with related publication experience. Moreover, 10 students who were game players were engaged in matching the items to the affordances. Changes were then made to improve wording. A pretest was conducted to filter out items that had significant cross-loadings.

The sample items of the two affordances were as follows: "This online game offers me the possibility to understand the mechanism behind the game" (mechanism affordance) and "This online game offers me the possibility to role-play my character in the world of the game" (role-playing affordance). We also used system-captured data as a measure of skill level. Specifically, we used lol.moa.tw to check each respondent's officially recognized skill level, which ranged from 0 (nonleveled) to 9 (challenger).

3.2.2. Second-wave research variables

Our study assessed goal progress by adapting the measures of Teng (2017). One item was "I am progressing toward the goals I want to attain in this online game". We measured concentration by modifying the methods of Agarwal and Karahanna (2000). An sample item was "When playing this online game, I experience full concentration ___% of the time." These items were assessed using a behavioral-continuous scale, reducing concerns of common method variance (CMV) (Sharma et al., 2009).

3.2.3. Third-wave: four-month follow-up gameplay extension

We measured gameplay extension by using respondents' account names to manually check the number of months out of four during which a player continued to play the game (LoL) from the end of the second data survey.

3.2.4. Control variables

We included six control variables to rule out their influences. As suggested by the literature (Shiau et al., 2024), the control variable choices are based on past studies showing their relevance: gender (Flavián et al., 2024; Hussain et al., 2025), age (Flavián et al., 2024; Hussain et al., 2025), education (Hussain et al., 2025), income (Pham et al., 2024), years spent using the game (Pham et al., 2024), and weekly hours spent using the game (Pham et al., 2024). These control variables are straightforward. Hence, it is appropriate to assess each of them using a single item. Single items can be accepted in the literature (Jo, 2025; Ma et al., 2025).

The scales for the two affordances and goal progress had response options that ranged from 1 (strong disagreement) to 5 (strong agreement). The skill level and gameplay extension were taken from system-captured data; therefore, they should be objective, precise, and not have measurement errors, justifying their single-item measurement.

3.3. Addressing common method variance by research design

This research has several methodological merits that help prevent CMV. First, it collects data from multiple sources, fully negating CMV (Podsakoff et al., 2003). Second, data were gathered using two waves of surveys and system-captured data for four months. In total, the data spanned more than six months, creating a significant temporal sequence of the constructs and therefore reducing CMV (Podsakoff et al., 2003). Third, this study used both system-captured data and behavioral-continuous scales that can effectively address CMV concerns

(Sharma et al., 2009).

3.4. Statistical properties

For consistency with past studies (Huang et al., 2025a; Yu et al., 2025), we examined the factor structure underlying the measures by computing loadings and cross-loadings. We observed no significant cross-loadings, as shown in Table 2.

All the constructs in our study items had Cronbach's α values $\geq .84$, average variance extracted (AVE) values $\geq .70$, and composite reliability (CR) values $\geq .88$, reflecting adequate reliability. The indicator loadings were $\geq .77$, indicating convergent validity. All correlations were less than the positive square roots of the AVE, demonstrating adequate discriminant validity. Following the literature (Hussain et al., 2025; Jo, 2025; Yu et al., 2025), we also used the heterotrait–monotrait (HTMT) ratios to examine discriminant validity. Appendix B shows the HTMT ratios, all of which were less than .85, indicating adequate discriminant validity. The study measures fit the data properly: IFI = .98, CFI = .98, NNFI = .97, and RMSEA = .05.

3.5. Testing common method variance

Table 3 lists the correlations among the research concepts, which ranged from .03 to .57. These magnitudes may not reflect significant common method variance (CMV). We also tested the CMV. First, this research followed the literature (Bian & Yang, 2025; Jo, 2025; Yu et al., 2025), and used Harman's test. We found that the first factor explained only 31% of the data variance. Hence, CMV should not be significant. Second, per the literature (Malhotra et al., 2006), we used the second-smallest correlation (i.e., .002) as the CMV. This is negligible. Third, we followed the literature (Lindell & Whitney, 2001) by applying CMV to modify the correlations. The significant correlations remained significant. Thus, CMV would not be substantial.

4. Results

4.1. Participants' profile

Most of the 379 participants were men (87.9%), which is in line with the sampling pool: 87% of LoL players were men (Priori Data, 2025). Most participants were < 30 years old (82.0%), had received a college/university education or higher (93.9%), and had an income below NT\$900,000 (94.7%). Moreover, many respondents reported playing the nominated game for < 21 h a week (87.3%) or for five years or more (77.8%). Table 4 presents our participants' profile.

Table 2
Loadings and Cross-Loadings.

	1	2	3	4	5	6
Mechanism Affordance–1	.86	.20	.07	.08	.13	–.09
Mechanism Affordance–2	.85	.13	.06	.13	.01	.11
Mechanism Affordance–3	.85	.08	.13	.19	–.07	.05
Role-playing Affordance–1	.07	.87	–.00	.12	–.09	.12
Role-playing Affordance–2	.13	.85	.09	.07	.08	–.09
Role-playing Affordance–3	.20	.84	.02	.17	.03	–.03
Concentration–1	.10	–.01	.89	.13	.02	.08
Concentration–2	.07	.04	.94	.13	–.02	.05
Concentration–3	.07	.07	.87	.10	.05	–.05
Goal Progress–1	.12	.15	.10	.88	.09	.00
Goal Progress–2	.13	.12	.12	.90	.00	.07
Goal Progress–3	.15	.09	.16	.88	.03	.01
Skill Level*	.05	.02	.03	.10	.92	.32
Gameplay Extension*	.06	.00	.07	.06	.32	.92

Note. * denotes objective, system-captured data.

Table 3
Statistics of the Study Constructs.

	1	2	3	4	5	6
1. Mechanism Affordance	.84					
2. Role-playing Affordance	.33*	.84				
3. Concentration	.21*	.11*	.87			
4. Goal Progress	.32*	.29*	.29*	.91		
5. Skill Level	.10*	.05	.07	.15*	NA	
6. Gameplay Extension	.11*	.03	.10*	.12*	.57*	NA
Mean	3.26	2.62	65.76	3.27	1.93	1.44
Standard Deviation	0.93	0.99	21.51	0.83	1.99	1.59
Cronbach's α	.85	.84	.88	.90	NA	NA
Composite Reliability (CR)	.88	.88	.90	.93	NA	NA
Average Variance Extracted (AVE)	.71	.70	.76	.82	NA	NA

Note. Bolded figures on the diagonal represent the positive square roots of the AVE values. NA denotes “not applicable.” * $p < .05$.

Table 4
Participants' Profile.

Feature	Category	No.	%
Gender	Male	333	87.9
	Female	46	12.1
Age	19–20 years	10	2.6
	21–25 years	157	41.4
	26–30 years	144	38.0
	31–35 years	54	14.3
	≥ 36 years	14	3.7
Education	High school or less	23	6.1
	University/college	279	73.6
	Graduate institute or above	77	20.3
Income	$\leq$ NT\$300,000	198	52.2
	NT\$300,001–600,000	118	31.1
	NT\$600,001–900,000	43	11.4
	$\geq$ NT\$900,001	20	5.3
Gameplay Years	< 1 year	33	8.7
	≥ 1 year and < 5 years	51	13.5
	≥ 5 years and < 10 years	256	67.5
	≥ 10 years	39	10.3
Gameplay Hours per Week	< 7 h	129	34.0
	≥ 7 and < 14 h	117	30.9
	≥ 14 and < 21 h	85	22.4
	≥ 21 and < 28 h	23	6.1
	≥ 28 h	25	6.6

Note. No.=number. %=percentage.

4.2. Testing the hypotheses

LISREL software was employed to conduct structural equation modeling (SEM) to test our research hypotheses. Orthogonalizing methods were used to test the moderation hypotheses (Little et al., 2006). Table 5 and Fig. 2 present the test results. The study hypotheses were mostly supported. The role-playing affordance was not associated with concentration; thus, H2a was not supported. A possible explanation is that many players may not enjoy becoming engaged in backstories of the game characters; thus, the action possibility of role-playing the characters may not attract much of their attention or concentration. Goal progress was not associated with gameplay extension; thus, H4 was not supported. We speculate that greater goal progress implies becoming closer to a goal. After attaining a goal, a player may experience cognitive closure, which offers an opportunity to reconsider gameplay continuance, affecting gameplay extension. Mechanism affordance and role-playing affordance did not directly fuel gameplay extension, not supporting H5a and H5b. The reasons should be that the affordances require users to actualize such affordances. Hence, mediators or moderators (as in Fig. 1) are needed. Skill level did not moderate the link between mechanism affordance and gameplay extension; thus, H6a was not supported. A possible explanation is that enhanced mechanism affordance may constantly motivate players to devise optimal strategies

Table 5
Results of Hypothesis Testing.

	IV	DV	Path	p-value	Result
H1a	Mechanism Affordance	Concentration	.25	< .001	Supported
H1b	Role-playing Affordance	Concentration	.05	.401	Not Supported
H2a	Mechanism Affordance	Goal Progress	.31	< .001	Supported
H2b	Role-playing Affordance	Goal Progress	.21	< .001	Supported
H3	Concentration	Gameplay Extension	.09	.045	Supported
H4	Goal Progress	Gameplay Extension	-.01	.826	Not Supported
H5a	Mechanism Affordance	Gameplay Extension	.07	.191	Not Supported
H5b	Role-playing Affordance	Gameplay Extension	-.06	.255	Not Supported
H6a	Mechanism Affordance × Skill Level	Gameplay Extension	.09	.067	Not Supported
H6b	Role-playing Affordance × Skill Level	Gameplay Extension	-.11	.022	Supported
H7	Skill Level	Gameplay Extension	.59	< .001	Supported

Note. IV=independent variable. DV=dependent variable. Path=standardized path coefficient.

and experiment them, regardless of players' skill levels.

Note. Dotted lines are the paths possessing coefficients that are insignificant. $*p < .05$. Items in ellipses were assessed using survey responses. Rectangles denote concepts measured using system-captured data. Control variables were modeled to have effects on all endogenous variables. To increase clarity, the impacts of control variables are not included in the figure.

Our theoretical model provides a sufficient fit of the data: IFI= .96, CFI= .96, NNFI= .95, and RMSEA= .045. We also found that the structural model significantly explained the variance in each endogenous variable, i.e., 14% concentration, 24% goal progress, and 41% gameplay extension. These proportions represent effect sizes magnitudes of medium to large (Cohen, 1992).

4.2.1. Testing reverse causality

We tested and verified that reverse causality is unlikely to affect our

results. Specifically, the hypotheses were reversed in their directions. The reversal boosted the statistical unfit, i.e., $\Delta df = 40$, $\Delta \chi^2 = 145.11 > \chi^2(40) = 55.76$, providing statistical evidence that did not support the presence of reverse causality.

4.3. Robustness checks

4.3.1. Control variables and their impacts

Women reported higher scores in concentration. Moreover, age was positively associated with concentration, whereas education was positively associated with goal progress. Both years spent using and hours per week spent using the nominated game were positively associated with concentration, whereas hours per week were also positively related to goal progress. These findings offer managerial insights, but their theoretical underpinnings may provide opportunities for future studies.

4.3.2. Sensitivity analysis

Notably, the inclusion of control variables may have altered our research results. Thus, we attempted to remove the control variables to examine whether excluding the control variables may affect the analytical results. This exclusion did not alter the analytical results. The path coefficients were affected by a minor magnitude, i.e., $\leq .03$.

4.3.3. Testing endogeneity and reverse causality

It may be questioned whether increased gameplay extension enhances gaming skills. Our study adopted a correct temporal data sequence, i.e., skill first and gameplay extension second. We further tested whether gaming skill is an endogenous variable, i.e., gameplay extension increases gaming skill. This encouraged us to reverse the path from skill to gameplay extension. The reversal significantly deteriorated the statistical fit, i.e., $\Delta df = 10$, $\Delta \chi^2 = 21.57 > \chi^2(10) = 18.31$. In short, gaming skill should be an exogenous, not an endogenous, variable in our model.

5. Discussion

Overall, this study initiated a conceptualization of goal pursuit using concentration and goal progress. Moreover, the findings indicate that concentration has a stronger effect than does goal progress on gameplay extension. This finding supports the GGT by showing that concentration is the essential element in the theoretical process of goal pursuit. The research findings answer the question of how game affordances can be designed to extend players' actual gameplay. The mechanism affordance can exert a stronger effect than the role-playing affordance. These

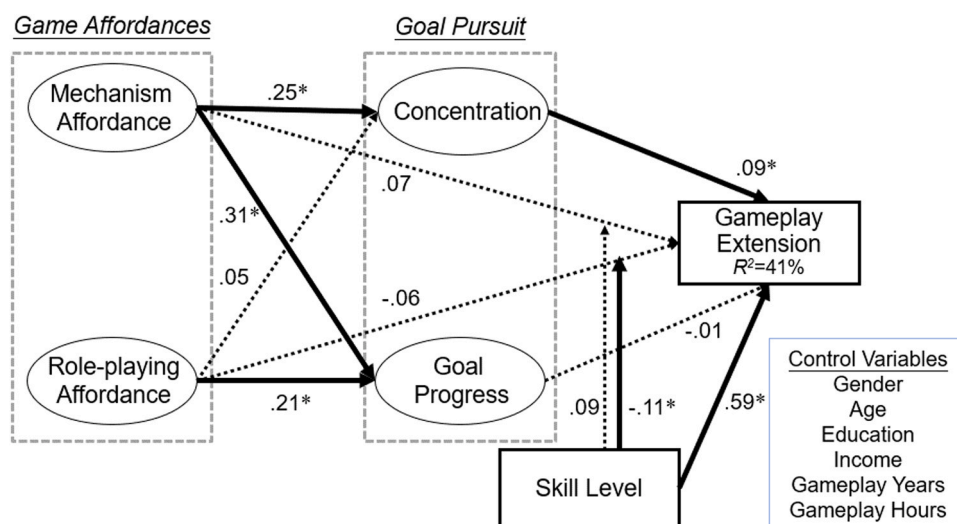


Fig. 2. Testing Results.

findings offer clear implications for game providers.

5.1. Implications for theory

5.1.1. Contribution to game research

The literature on gameplay extension has examined various antecedents of further gameplay: flow (Rahman et al., 2024), reward responsiveness, fun seeking (Aydinliyurt et al., 2021), ease of imagination (Flavián et al., 2024), and cooperation knowledge (Bian & Yang, 2025). Compared with this literature, our study presents a novel decomposition of goal pursuit in games into concentration and goal progress. Such a new decomposition generates a typology that future studies can adopt and further examine its role in formulating gameplay behavior.

Past game research has shown that gaming skills fuel players' loyalty (Bian & Yang, 2025). Comparatively, our findings add the boundary condition that skill level *weakens* the impact of role-playing affordance. This finding allows future game research to include players' skill levels that may change the impact of game design.

Past game research has also revealed that goal pursuit is vital for player retention (Huang et al., 2022; Teng et al., 2022); however, our study extends such research by demonstrating that players could be retained for at least four months. This illustration of gameplay extension requires further research to explore whether the retention period could be further extended.

Research on gaming avatars has revealed that the use of avatars can increase players' identification with avatars and player groups (Tai et al., 2025; Teng et al., 2023) and their purchase intention of virtual items (Ahn et al., 2024). Compared with these studies, the present study indicates for the first time that the use of gaming avatars (e.g., role-playing affordances) can also help enhance players' concentration and goal progress.

We uniquely measured gameplay extension in terms of the number of months a player actually continued playing a game. This new initiative shows its methodological contribution to the game research stream.

5.1.2. Contribution to goal gradient theory

The GGT literature posits that the closer a goal becomes, the larger it appears (Förster et al., 1998), i.e., GGT posits an acceleration process: individuals are motivated to exert greater effort, or accelerate their effort exertion, as they approach the goal (Kivetz et al., 2006). As we found that goal progress was not significantly related to gameplay extension (a relative long time period), we did not find evidence supporting that the acceleration process of GGT has a substantiated effect on long-term engagement (in term of gameplay extension). This may be because that momentum boosting acceleration may cease immediately when the goal is achieved. Moreover, acceleration may take much energy and therefore cannot last for long, comprising a boundary condition of the acceleration process of GGT.

Past GGT research has indicated that two elements—goal proximity motivation—are used for attaining in-game goals (Huang et al., 2022; Teng et al., 2022). These two elements are important but do not fully capture the essence of the GGT: greater effort toward achieving goals (Kivetz et al., 2006). The present study fills this gap by uniquely incorporating the concept of concentration and revealing its strong effect on player retention. This finding expands the GGT by showing that concentration should be essential to the theoretical process of the GGT.

Compared with the GGT literature on games (Huang et al., 2022; Teng et al., 2022), the present study is new in that it proposes two novel and effective affordances: mechanism and role-playing affordances. By introducing the two new affordances, our research revealed new triggers that facilitate goal pursuit in the context of online gaming, enhancing the predictive premises of GGT.

5.1.3. Contribution to affordance research

The information systems literature on affordances has clarified that

affordance perceptions lead to affordance actualization, thus generating effects (Strauss et al., 2024), including positive user responses (Tarafdar et al., 2024) and mood modifications (Lee et al., 2021). Our study advanced affordance research by proposing two affordances that fuel two components of goal pursuit. These findings lay the foundation for future affordance research examining goal pursuit.

5.2. Implications for practice

Many online games contain mechanism affordances and role-playing designs that can be used strategically to influence the goal process, further prolonging gameplay extension. All these are the main study concepts of our research model; therefore, our findings could be applied in many online games. For example, another game, PUBG: Battlegrounds, records and displays players' win rates; Arena of Valor enables players to choose different roles; and Free Fire encourages players to communicate in real time to achieve in-game victory. All these methods share the same features of LoL and conform to our research model. Therefore, our findings have enhanced generalizability.

We found that the mechanism affordance can influence players' concentration, increasing gameplay extension. Hence, game makers could hold related events, e.g., game launch parties, and record skillful players' in-game practices to demonstrate the game mechanisms. In so doing, players could be inspired to learn and implement the best game strategy to increase their opportunities to achieve in-game victory. Hence, the events could motivate players to concentrate on researching the game mechanism and increasing their gameplay extension.

We found that the role-playing affordance could enhance player-perceived goal progress. Hence, game makers could increase the number of characters available for role-playing at the beginning of the gameplay. Players could then choose occupations for role-playing, such as fighters or warlocks, and enjoy using their special skills (e.g., critical hits or casting spells) to advance their progress toward reaching the in-game goals. Additionally, game makers could record the use history of players' characters, e.g., favorite characters or win rates for each character, to clearly understand how their role-playing is associated with progress toward their in-game goals.

We found that the skill level *weakened* the impact of role-playing affordance on gameplay extension. This indicates a contingent effect, i.e., the role-playing affordance may distract highly skilled players from displaying their superior skills to play the game longer. Hence, we suggest that game makers not intentionally promote the role-playing affordance in skill-demanding games (e.g., real-time strategy games or action adventure games) for players at higher game levels.

5.3. Limitations and future research directions

As its focal context, this study chose the highly popular online game LoL, which is characterized by five-versus-five team competitions. Hence, it is suitable for examining players' skills and goal pursuits. However, the study concepts used in our model are also applicable to the majority of online games. Hence, future research could extend this model by adding the unique features of other online games.

Many games prepare goals for players, justifying the generalizability of our findings to games that include possible goals for players. However, not all players assume the goals prepared by the game. Some players may simply seek fun or check what will happen in games, comprising a research limitation for our research. Future studies could apply theories that *exclude* goals to explain what if players simply seek fun, collectively with our research, to form a complete picture for explaining gameplay.

This study focuses only on PvP (player versus player) games, in which players compete with each other to gain a reward if they win. In contrast, PvE (player versus environment) games do not include competitions among players, which is fundamentally different from the PvP context in this study (i.e., LoL). Future studies can examine whether our

findings apply to PvE games and what changes would make the findings better fit PvE games.

We chose month-based measures for gameplay extension because months are long enough and aligned with monthly calendars. Hence, months are among the suitable units (weeks, months, and quarters) when assessing business performance (Murphy, 2025). Future studies can replicate our research and investigate how to increase gameplay extension across quarters. Moreover, the total number of game sessions can be used for assessing gameplay intensity in a specific, short time period (e.g., a week), whereas our focus is to retain players for a longer time.

This study examined the mechanism and role-playing affordances and whether players' skill levels moderated the relationship between the two affordances and gameplay extension. A single study cannot include all factors, and our study is no exception. Hence, future research could explore other possible factors, such as the sources or process variables that affect gameplay extension.

Novel technologies, such as artificial intelligence and virtual reality, enable new action possibilities. Future research could also examine whether context-specific elements in such technologies may change the results of our research model.

Concentration indicates increased cognitive effort (Förster et al., 1998). However, when players feel that a game is too difficult, they may give up. We cannot exclude this possibility; however, games should have incentives to retain their players and avoid excessively difficult designs that might scare away their players. Hence, extreme cases in which a game is perceived as being too difficult may require that future studies examine their unfavorable effects.

6. Conclusions

The competitive gaming industry compels companies to strive to

extend players' actual gameplay. Without our research, game providers might mistakenly view concentration and goal progress as having the same importance. Our findings show that concentration should be more important than goal progress in extending gameplay, advancing our understanding of goal gradient theory. Our findings also clarify that game affordances may not similarly drive gameplay extension for all players. Specifically, role-playing affordances may not work well for highly skilled players. In summary, our findings provide practical insights for game providers that the mechanism affordance can fuel goal pursuit and, therefore, help extend players' actual gameplay.

CRedit authorship contribution statement

Alan R. Dennis: Writing – review & editing, Validation, Supervision, Conceptualization. **Hsin-Yu Hung:** Writing – original draft, Visualization, Investigation, Data curation. **Tzu-Ling Huang:** Writing – original draft, Methodology, Formal analysis, Conceptualization. **Gen-Yih Liao:** Writing – original draft, Investigation, Funding acquisition, Conceptualization. **Ching-I Teng:** Writing – review & editing, Writing – original draft, Project administration, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors have no competing interests to declare.

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Appendix A. Scale Items and Statistics

Construct	Item	<i>M</i>	<i>SD</i>	λ
Mechanism Affordance	This online game offers me the possibility to understand the mechanism behind the game.	3.23	1.07	.85
	This online game offers me the possibility to know the precise numbers and percentages underlying the game mechanics.	3.23	1.08	.82
	This online game offers me the possibility to analyze the game rules.	3.31	1.02	.85
Role-playing Affordance	This online game offers me the possibility to experience my character's stories in the world of the game.	2.52	1.11	.82
	This online game offers me the possibility to role-play my character in the world of the game.	2.73	1.14	.81
	This online game offers me the possibility to use my character to experience the fantasy world in person in the game.	2.61	1.14	.88
Concentration	When playing this online game, I experience full concentration __% of the time.	72.64	18.75	.85
	When I play this online game, I am fully focused on (or absorbed in) the game __% of the time.	66.17	23.84	.98
	When playing this online game, nothing distracts me for __% of the time.	58.47	28.13	.77
Goal Progress	I am progressing toward the goals I want to attain in this online game.	3.35	0.89	.89
	I am closer than before to attaining the goals I set in this online game.	3.25	0.93	.92
	Overall, I have made substantial progress toward attaining the goals I set in this online game.	3.21	0.92	.91
Skill Level*	Officially recognized skill level in year 2020	1.93	1.99	NA
Gameplay Extension*	The number of months out of four in which a player continues to play the game from a particular time point	1.44	1.59	NA

Note. *M*=mean; *SD*=standard deviation; λ =indicator loading; NA=not applicable; *=objective, system-captured data.

Appendix B. Heterotrait-Monotrait (HTMT) Ratios

	1	2	3	4
1. Mechanism Affordance				
2. Role-Playing Affordance	.39			
3. Concentration	.22	.12		
4. Goal Progress	.36	.34	.31	

Note. Skill level and gameplay extension were represented by single indicators that were objective and system-captured, and therefore not included for computing HTMT ratios.

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